

captured a size effect (small companies outperform large companies) and a value-growth effect (value stocks do better than growth stocks). They labeled their size factor “*SMB*,” for small stocks minus big stocks, and their value-growth factor “*HML*,” for high book-to-market stocks minus low book-to-market stocks. Value stocks are stocks with depressed prices, and the *HML* factor normalizes these prices (market value) relative to book value. Hence value stocks are stocks with high book-to-market ratios. Chapter 7 covers the Fama–French model and the economics behind the size and value premiums. (Although the size effect is much weaker today, it is still instructive to include *SMB* to explain some of Berkshire Hathaway’s performance, as I show below.)

The *SMB* and *HML* factors are long–short factors. They are mimicking portfolios that consist of simultaneous \$1 long and \$1 short positions in different stocks. That is,

$$SMB = \underbrace{\$1 \text{ in small caps}}_{\text{Long}} - \underbrace{\$1 \text{ in large caps}}_{\text{Short}}$$

and so *SMB* is designed to capture the outperformance of small companies versus large companies. The *HML* factor picks up the outperformance of value stocks versus growth stocks:

$$HML = \underbrace{\$1 \text{ in value stocks}}_{\text{Long}} - \underbrace{\$1 \text{ in growth stocks}}_{\text{Short}}.$$

The Fama–French benchmark holds positions in the *SMB* and *HML* factor portfolios, along with a position in the market portfolio as in the traditional CAPM.

Fama and French extended the CAPM regression in equation (10.9) to include a size factor and a value factor. The Fama–French benchmark can be estimated by running the following regression:

$$r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + sSMB_t + hHML_t + \varepsilon_{it}, \quad (10.10)$$

which adds the *SMB* and *HML* factors to the standard market factor.

The *SMB* and *HML* factor loadings are given by s and h , respectively. If a stock co-moves neither with small nor large stocks, it’s a medium-size stock, and s would be zero. As it starts moving with small stocks s becomes positive, and if it moves together with large stocks, s is negative. Likewise, h measures how much a stock is acting like other value stocks: positive h s indicate the stock has a value orientation, and negative h s indicate the stock is acting like a growth stock. The market itself is neither small nor big and neither value nor growth, so the market has zero s and h loadings.

Estimating the Fama–French regression for Berkshire Hathaway yields the following coefficients: